

SUICIDAL TENDENCIES DETECTION USING CNN AND RANDOM FOREST ALGORITHM



Major Project submitted in partial fulfillment of the requirement for the award of the
degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

Under the esteemed guidance of

Mrs.V. Sravanthi
Assistant Professor

By

Lavudya Revanth (21R11A05H4)
Sende Sai Kumar (21R11A05K6)
Gujja Mokshith (21R11A05G5)



Department of Computer Science and Engineering
Accredited by NBA

Geethanjali College of Engineering and Technology
(UGC Autonomous)

(Affiliated to J.N.T.U.H, Approved by AICTE, New Delhi)
Cheeryal (V), Keesara (M), Medchal.Dist.-501 301.

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Geethanjali College of Engineering & Technology
(UGC Autonomous)

(Affiliated to JNTUH, Approved by AICTE, New Delhi)
Cheeryal (V), Keesara(M), Medchal Dist.-501 301.

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This is to certify that the B. Tech Mini Project report entitled “**SUICIDAL TENDENCIES DETECTION USING CNN AND RANDOM FOREST ALGORITHM**” is a bonafide work done by **Lavudya Revanth (21R11A05H4)**, **Sende Sai Kumar (21R11A05K6)**, **Gujja Mokshith (21R11A05G5)**, in partial fulfillment of the requirement of the award for the degree of Bachelor of Technology in “**Computer Science and Engineering**” from Jawaharlal Nehru Technological University, Hyderabad during the year 2024-2025

Internal Guide

HOD - CSE

Mrs.V.Sravanthi
SreeLakshmi
Professor

Dr A
Assistant
Professor

External Examiner

Geethanjali College of Engineering & Technology
(UGC Autonomous)

(Affiliated to JNTUH Approved by AICTE, New Delhi)

Cheeryal (V), Keesara(M), Medchal Dist.-501 301.

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DECLARATION BY THE CANDIDATE

We, **Lavudya Revanth, Sende Sai Kumar, Gujja Mokshith** bearing Roll Nos. **21R11A05H4, 21R11A05K6, 21R11A05G5**, hereby declare that the project report entitled **“SUICIDAL TENDENCIES DETECTION USING CNN AND RANDOM FOREST ALGORITHM”** is done under the guidance of **Mrs. V. Sravanthi, Assistant Professor**, Department of Computer Science and Engineering, Geethanjali College of Engineering and Technology, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

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**Lavudya Revanth (21R11A05H4),
Sende Sai Kumar (21R11A05K6),
Gujja Mokshith(21R11A05G5),
Department of CSE,
Geethanjali College of Engineering and
Technology,
Cheeryal.**

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ABSTRACT

This project presents a multifaceted approach to detecting suicidal tendencies by integrating Human-Computer Interaction, Natural Language Processing, and voice pattern analysis. The system captures and analyzes facial gestures, speech patterns, and messaging behaviours to identify early signs of suicidal intent. By employing machine learning techniques such as Convolutional Neural Networks (CNN) and Random Forest, the model processes data from various sources, including social media and messaging platforms. The use of a correlation matrix ensures reliable predictions across multiple data types, significantly improving detection accuracy compared to existing single-method approaches. This system aims to provide timely alerts to family and healthcare professionals, potentially preventing suicidal actions.

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1. INTRODUCTION

1.1. About the project

This project aims to focus on the people who have an intension to kill themselves. In this consequence, multi-faceted method that can detect this tendency and intimate the family, friends or the close ones beforehand can prove to be a boon for the invention. This project tends to consider an electronic device particularly a mobile (as used by most of them) is the key element. This device is used to capture different elements like facial gestures, speech recognition and many more. A trivial concept of incorporating different aspects like: Facial Gestures, Voice Recognition and Messaging Patterns follow the bandwagon along with the technical biproducts of the project. Facial gestures include unhappy expressions like sad, dull, tired; voice patterns include low voices that sound dull are easy to recognize that someone is unhappy; Texting patterns include unusual texting patterns that indicate lack of interest in doing activities. It frequently co-occurs with anxiety or other psychological and physical illnesses and affects how those afflicted feel and behave.

According to the WHO study, there are 322 million people estimated to suffer from depression, equivalent to 4.4% of the global population. Nearly half of the in-risk individuals live in the South-East Asia (27%) and Western Pacific region (27%) including China and India. In many nations depression is still under-diagnosed and left without any effective therapy that can result into an extreme self that can ultimately lead to suicide. Suicide represents an important social issue. In accordance with the World Health Organization (WHO), nearly 700,000 million individuals worldwide commit suicide each year, and many more, particularly those in their twenties and thirties, attempt suicide. Suicide is the second most common cause of death in people aged 10 to 34. Suicidal tendency, as well referred as suicidal ideation, refers to when a person thinks about self-harm. Suicidal ideation can impact individuals of every age for a wide range of reasons along with shock, anxiety, guilt, stress, and depression. Longterm depression can result in suicide if sufficient treatment is not sought, despite the reality that a large percentage of people who have suicidal thoughts do not attempt suicide. Medical professionals and pharmaceuticals can be used to reduce a person's predisposition towards suicide.

However, because of the negative stigma connected with medical treatments, most people who have suicidal thoughts avoid them. This project intends to focus on people who intend to commit suicide. As a result, a multifaceted technique which can detect this tendency and notify family, friends, or close ones ahead of time can be a boon to the invention.

The key element in this project is an electronic device, specifically a mobile phone (as used by the majority of them). This device is used to capture various elements such as facial gestures, speech recognition, and many others. A simple concept of incorporating various aspects such as Facial Gestures, Voice Recognition, and Messaging Patterns joins the bandwagon alongside the project's technical by-products. Unhappy facial expressions such as sad, dull, and tired are easy to recognise; voice patterns such as low voices that sound dull are easy to identify that someone is unhappy; Text messaging patterns that are unusual indicate a lack of interest in doing activities. According to one study, 78% of hospital inpatients who committed suicide denied having suicidal thoughts during their last verbal communication.

Thus, A novel, data-driven tool for analysing acute suicide risk is urgently needed. People have to anticipate not only who is at a greater chance of committing suicide in general, but also the time that person is at a higher risk. The increased use of smartphones and information services such as email, blogs, crowd-sourcing sites, and social media has resulted in an increase in the volume of unstructured text data. Text mining techniques applied to person-generated data, such as text may reveal how communication patterns and media use change as an individual's risk state rises. (For example, from depression to suicidal tendency to suicide attempt). As per Pew Research Centre, in the United States, 99% of Millennials are using the online services and 92% already have a smartphone. Millennials, the first generation to be engrossed in the world of social media and technology, are indeed an extremely vulnerable population, as selfharm is the leading mortality rate for individuals aged 15 to 34.

1.2. Objective

The objective of this system is that it is capable of detecting suicidal tendency in a specific person. Unlike other existing system this system can ensure the ability to focus on different technical aspects rather than single one as the output is not reliable in earlier systems as well as it was not practically possible to indicate a clear demarcation in only one aspect of implementation.

The project aims to find co-relation between the three components present below.

1. Facial Gesture Detection – Human Computer Interaction
2. Speech Recognition – Natural Language Processing
3. Messaging Patterns – Text Tokenization through NL

2. SYSTEM ANALYSIS

2.1. Existing System

Since suicide is not a decision made up in a single day, many researches had been done in the past on various aspects. While some of them included how suicides occur and others concluded with some methodologies on how to detect suicidal intension. Distinct approaches using Human Computer Interaction, Natural Language Processing and using Convolutional Neural Networks were highly popular. But these systems possessed a drawback that, it was not practically possible to indicate a clear demarcation in only one aspect of implementation. In course of time, numerous approaches have been proposed with the advancement in the technology which is to perform text mining and sentiment analysis on social media platforms like reddit and twitter.

Disadvantages:

- Can predict outcome only from one aspect.
- The output is not reliable

2.2. Proposed System

This project aims to propose a system that is capable of detecting suicidal tendency using multiple approaches. Three technologies namely Human Computer Interaction, Natural Language Processing and voice pattern analysis are taken into consideration to perform a detailed execution. Later a correlation matrix is proposed to be implemented that is able to find strong or weak correlations between the above three components. The goal of this study is to suggest a system that can identify suicidal tendencies using a variety of methods. To carry out a thorough execution, three technologies—Human Computer Interaction, Natural Language Processing, and voice pattern analysis—are taken into account.

The implementation of a correlation matrix that can identify strong or weak correlations between the aforementioned three components is later suggested. It has the advantages of predicting outcomes from a variety of perspectives and providing accurate results thanks to a correlation matrix. The goal of this study is to show a fully functional device that can identify a suicide intention. The spectrum of intended outputs includes 5 recording facial expressions and identifying the victim's utterances. We use Human Computer Interactions for Facial Gesture Detection, Natural Language Processing for Speech Recognition, and Text Tokenization via NL for Messaging Patterns. Text Pattern Recognition - TWINT is used to prepare the dataset before analysing non depressive data. These two sets of data are combined. Torch Text is used for tokenization. Speech Pattern Recognition - Use the DAIC-WOZ dataset to separate audio signals into spectrograms. Create a CNN architecture and recognise lower frequencies patterns with CNN and SVM. Facial Gesture Recognition - Use the FER2013 dataset to classify emotions using CNN architecture.

Separate positive as well as negative emotions now. Use CNN on negative emotions to detect suicidal intent. Text analysis (TA), a machine learning technique, is used to automatically extract helpful information from unstructured text input. Businesses utilise text analysis technologies to quickly absorb papers and web data and turn them into insightful information. Text analysis allows you to extract precise information from tens of thousands of emails, like names, company names, or keywords, and arrange survey responses by attitude and topic. Using specifically created algorithms, sound recognition (and voice recognition) recognises patterns in audio signals. The algorithm then selects and groups data based on predetermined criteria or by using common parts and components.

The algorithm also provides the opportunity for learning and improvement, which is essential for machine learning technology to produce results that are ever-more accurate. The name itself provides the simplest explanation: it is used to identify sound patterns in collections of audio data. patterns that use a series of flat lines and spikes to convey stories about the data. As a computer-based technology, the facial expression recognition system uses algorithms to instantly detect faces, decode facial expressions, and identify emotional states. It accomplishes this by using computer-powered cameras mounted on

computer screens, incorporated in laptops, smartphones, and digital signage systems, or cameras that are powered by batteries. Face detection, facial landmark detection, and facial expression and emotion classification are the general steps in facial analysis using computer-powered cameras.

Advantages:

- Predicts outcomes from multiple aspects.
- Reliable outcome due to correlation matrix.

2.3. Feasibility Study

2.3.1. Details

The feasibility of this project lies in analyzing the viability of the proposed system under real-world constraints. While all projects are theoretically feasible with unlimited resources and infinite time, practical limitations demand efficiency in resource utilization and adherence to timelines. Ensuring technical, operational, and economic feasibility is critical to securing project approval and successful implementation.

Key considerations in this analysis include:

Technical Feasibility

This involves evaluating whether the proposed system can integrate seamlessly with existing technologies. The project leverages tools like Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), and facial recognition algorithms, which are widely supported and readily available. It relies on Python-based libraries like PyTorch, TensorFlow, and TorchText, ensuring compatibility with modern computing environments. Hardware requirements include standard processors and GPUs, which are easily accessible.

Operational Feasibility

The system is designed to be user-friendly, meeting client requirements outlined in the initial analysis phase. By utilizing familiar platforms such as smartphones and cloud-based applications, the solution minimizes training overhead. The design ensures that end-users, such as mental health professionals and patients, can effectively interact with the system without specialized technical expertise.

Economic Feasibility

Cost analysis focuses on the affordability of the system within budget constraints. The project uses open-source tools and datasets, reducing initial development costs. It requires moderate computing resources, eliminating the need for expensive infrastructure. This ensures a favorable cost-to-benefit ratio, as the system delivers high-value outputs like early suicide risk detection, potentially saving lives and reducing long-term societal costs.

2.3.2 Impact on Environment

The project focuses on leveraging existing hardware (e.g., smartphones and servers), thereby reducing the need for new manufacturing. By utilizing cloud storage and computing, the system minimizes physical waste. However, care must be taken to responsibly manage electronic waste generated from hardware upgrades or replacements.

2.3.3 Safety

The system prioritizes user safety by employing non-invasive detection methods such as analyzing facial gestures, speech patterns, and text behavior. Data privacy is ensured through robust encryption and compliance with privacy standards. The architecture includes safeguards against data breaches and misuse, fostering trust among users.

2.3.4 Ethics

The system adheres to ethical guidelines, particularly concerning the sensitive nature of suicidal tendencies. It respects privacy by ensuring data is collected and analyzed with informed consent. The AI components are trained to avoid bias and provide objective

analyses. Transparency in system operations and engagement with mental health professionals ensures ethical deployment.

2.3.5 Cost

The solution entails both initial and operational costs. Initial expenses include software development and integration of existing hardware and cloud services. Operational costs cover data processing, storage, and periodic updates. The use of open-source frameworks significantly reduces these costs, while the benefits of reducing suicide rates and enabling early interventions offer immense societal and economic value.

2.3.6 Type

This system integrates multiple technologies, making it an AI-driven solution for mental health. It combines AI techniques (CNNs for image and speech analysis), NLP for text processing, and behavioral analytics to predict suicidal tendencies. The system also functions as a digital surveillance and intervention tool in mental health care, providing real-time insights and alerts.

2.3.7 Standards

The project complies with established technical and ethical standards. It uses widely accepted frameworks like YOLO for detection and PyTorch for modeling. Data security and privacy protocols align with global regulations, such as GDPR, ensuring user confidentiality. Additionally, it adheres to machine learning best practices for accuracy and reliability, providing a robust solution for suicide prevention.

2.4. Scope of the Project

The "AI-Based Suicidal Tendency Detection System" project encompasses several critical aspects to provide a robust and innovative solution for identifying suicide risk in individuals.

i. Objective

To develop an advanced system leveraging artificial intelligence and natural language processing to detect suicidal tendencies in textual data. The project aims to enhance early intervention capabilities by analyzing written or spoken language to predict suicide risk, enabling timely support and prevention measures.

ii. Technological Components

The system employs Convolutional Neural Networks (CNNs) and Torch Text, powerful tools for natural language processing (NLP). These technologies process and analyze large volumes of text data, such as social media posts, chat messages, and blog entries, to detect indicators of suicidal tendencies. The model is trained on labeled datasets, distinguishing between text that suggests suicidal ideation and text that does not. Future integration of multimodal data, including audio and video recordings, can provide additional behavioral and emotional insights, further improving the model's accuracy.

iii. System Features

The system provides real-time detection capabilities, utilizing advanced machine learning algorithms to analyze and classify text data effectively. Features include secure data processing, user-friendly interfaces for mental health professionals to review predictions, and comprehensive reporting tools. Additionally, the system is designed to incorporate deep learning and reinforcement learning techniques to enhance pattern recognition and identify subtle risk factors that traditional methods may overlook.

iv. Deployment and Installation

Deployment involves data collection and preprocessing, which includes gathering text data from sources like social media and anonymized case studies. The system is trained

and calibrated using these datasets to ensure accurate predictions. Installation includes setting up the model on cloud-based platforms or local servers for easy access by stakeholders, such as mental health practitioners. The system also integrates with user-friendly dashboards, offering insights and alerts in a secure and ethical manner to support effective suicide prevention strategies.

2.5. System Configuration

i. Hardware Requirements

Processor Intel Core i5 or above

Memory (RAM) Minimum 16 GB

Storage 512 GB or above

ii. Software Requirements

Operating System Windows, Linux, or macOS

Database SQLite

Programming Language Python

Frameworks/Libraries PyTorch, TensorFlow, and TorchText

iii. Development Environment

IDE/Code Editor Visual Studio Code or PyCharm

Version Control Git for managing source code

3. LITERATURE OVERVIEW

Animal detection systems aim to prevent or reduce the frequency of animal-vehicle collisions, which can cause human injuries, fatalities, and damage to property. These systems are designed to identify wild animals before they reach the road. Traditionally, measures such as warning signs, wildlife warning reflectors, and wildlife fences have been used to alert drivers and keep animals away from roadways.

In modern applications, object detection technology plays a key role. Unlike traditional methods, these systems use advanced algorithms to identify animals in images or videos. One notable algorithm in this field is YoloV3 (You Only Look Once version 3), known for its effectiveness and speed in real-time object detection. In certain areas, wildlife fencing is complemented by various crossing structures to further mitigate the risk of animal-vehicle collisions.

1. TITLE: IOT BASED SMART AGRICULTURE- 2022

Authors: PROF.N. GONDACHWAR, DR.R.S. KAWITKAR, Electronics and Telecommunication, college of engineering Pune.

Findings: The sensors and microcontroller of three nodes are successful interfaced with raspberry Pi and wireless communication is achieved between various nodes. All observations and experimental test proves that project is a complete solution to field activities, Irrigation problems and storage problem using remote control smart robot irrigation system and smart warehouse management system respectively. Implementation of such system in field can definitely help to improve the crops and overall productions. When things like household appliances are connected to a network, they can work together in cooperation to provide the ideal service as a whole, not as a collection of independently working devices. This is useful for many of the real-world applications and services, and one would for example apply it to build a smart residence. Therefore, automated inspection becomes a popular way for substituting human inspection as machines are good at repeating a same task without fatigue until now most automated crop inspector machine are realize by using visual inspection.

2. TITLE: IOT BASED SMART CROP MONITORING IN FARM LAND- 2022

Authors: N.GOWTHAMAN, V.NANDINI, S.MITHRA, N.PRIYA– Assistant professor, Department of ECE, SNS college of Technology, Coimbatore.

TN- INDIA- UG Student Department - ECE, SNS college of Technology TN-INDIA.

Findings: In this paper we purposed a method for efficient crop monitoring for agriculture field. With the application for IOT the data can be stored and retrieved from anywhere in this purposed work, the sensor part is limited only for monitoring Crops. The agricultural farming is done by using human eye's where farm are protected by which is made of grass pot as a inspectors however, the accuracy of human infection is unstable due to fatigue, and it's more challenging for human eyes to detect brides attack on the crops. To avoid person usually use the anonymous or rope to protect Crops. Therefore, automated inspection becomes a popular way for substituting human inspection as machines are good at repeating a same task without fatigue until now most automated crop inspector machine are realize by using visual inspection systems.

3. ANIMAL REPELLENT SYSTEM FOR SMART FARMING USING AI AND DEEP LEARNING- 2022

Authors: Mathanraj s1, Akshay k2, Muthukrishnan R3

Agricultural farm safety is a well-being wished time these days. Then, to achieve this, a machine based on a dream is proposed and completed utilization of Python and OpenCV and fostered an Animal Repellent device to victory the creatures. The execution of the product required the design and improvement of a mind-boggling gadget for shrewd creature shock, which coordinates recently progressed programming added substances and permits perceiving the presence and types of creatures in genuine time and to stay away from crop harm because of the creatures. Founded on the classification of the creature recognized, the limit processing instrument executes its DCNN Animal notoriety model to find the aim, and assuming a creature is identified, it sends lower back a message to the Animal Repulsor module alongside the type of ultrasound to produce in sync with the class of the creature. The proposed CNN became assessed in the made creature data set. The general exhibitions had been gotten utilizing

an extraordinary assortment of tutoring previews and investigating pictures. The got trial results of the accomplished analyses show that the proposed CNN gives a decent acknowledgment charge for more scope of entering training photos (exactness of around 98%).

4. DESIGN, DEVELOPMENT AND EVALUATION OF AN INTELLIGENT ANIMAL REPELLING SYSTEM FOR CROP PROTECTION BASED ON EMBEDDED EDGE-AI- 2021

Authors: Davide Adami, Mike Ojo, Stefano Giordano

Edge computing has become crucial for real-time applications by bringing processing and storage closer to end devices, reducing latency and enhancing data security. This study focuses on a Smart Agriculture application designed to protect crops from ungulate attacks using virtual fences powered by computer vision and ultrasound emission. The innovative system employs a device that generates ultrasonic signals to repel ungulates, reducing crop damage. It detects and recognizes different ungulate species and emits tailored ultrasonic signals.

Given the rural environment's energy and connectivity limitations, the system is based on IoT platforms that balance performance, cost, and energy consumption. We tested various edge computing devices, such as Raspberry Pi (with and without a neural compute stick) and NVIDIA Jetson Nano, running object detection models (YOLO and Tiny-YOLO) to determine the best platform for animal recognition integrated with the ultrasound generator. Results show the feasibility of deploying animal detectors on energy-efficient edge devices without compromising accuracy or real-time performance. The study also provides a cost/performance analysis and temperature measurements for each platform, offering insights for optimizing the system. Finally, the combined technology supports farmers and agronomists in decision-making and management processes.

4. SYSTEM DESIGN

4.1. System Architecture

A System Architecture is the conceptual model that defines the structure, behavior, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviors of the system. A system architecture can consist of system components and the sub-systems developed, that will work together to implement the overall system. There have been efforts to formalize languages to describe system architecture, collectively these are called architecture description languages (ADLs).

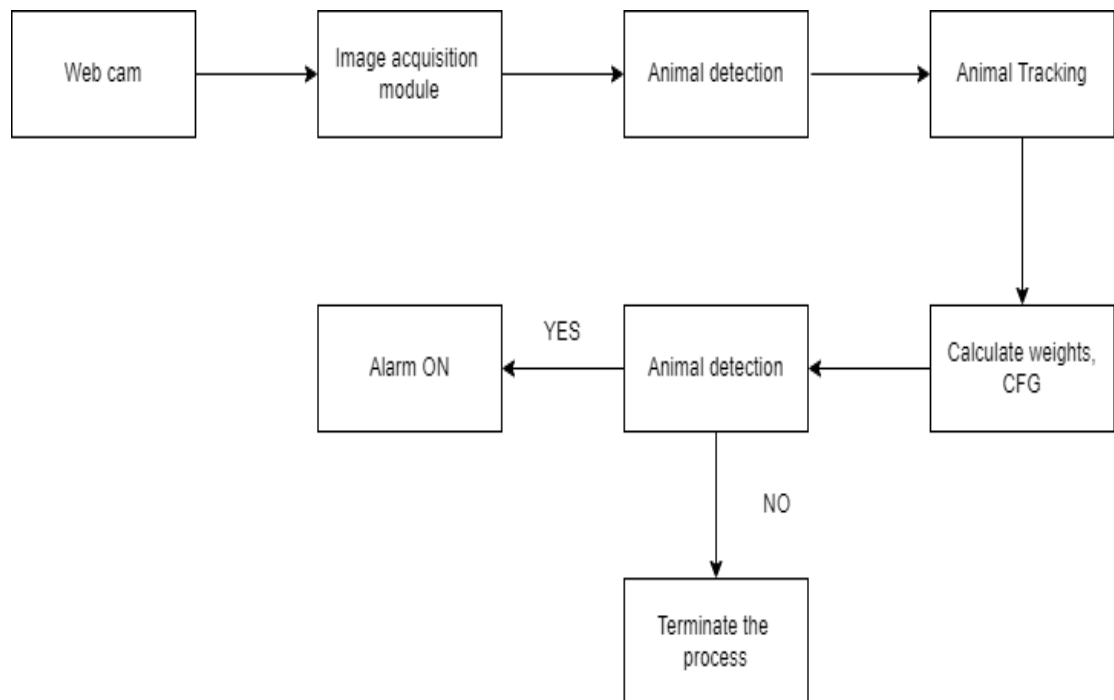


Fig 4.1 System Architecture

4.1.1. Module Description

1. User Authentication Module

A Login and Signup Module enables users to register and access an application securely. The signup process collects basic information like email, username, and password, often with email verification for added security. The login feature allows users to enter their credentials to access the application, ensuring their identity is authenticated.

2. Monitoring Module

In this project, we will monitor the entire farm at regular intervals using a camera that records the surroundings throughout the day. The system is designed to continuously capture video, ensuring comprehensive surveillance and real-time monitoring of the farm.

1. Detection Module

We are using the YOLOv3 algorithm, a real-time object detection technique, to identify animals in live video feeds. The captured video is divided into frames, with each frame analyzed for animal presence. If an animal is detected, it is identified by comparing with the coco.names dataset. This approach successfully performs animal detection and identification.

2. Alert Module

After identifying the animal using the YOLOv3 algorithm, a sound corresponding to the animal's name is played. This sound is designed to deter the animal and drive it away from the field. And alert message is sent to farmer.

4.2. UML Diagram

USE CASE DIAGRAM

A use case diagram in the Unified Modelling Language (UML) is a type of behavioural diagram defined by and created from a Use-case analysis. A use case diagram is used to represent the dynamic behavior of a system. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

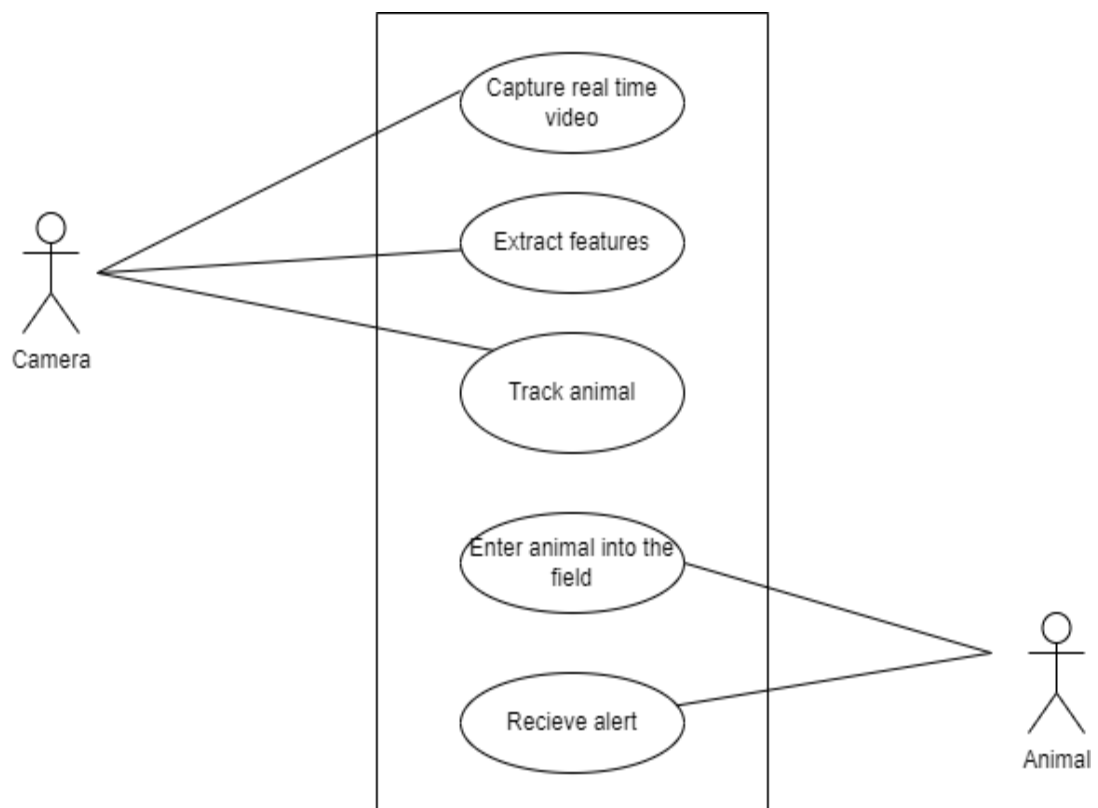


Fig 4.2 Use Case diagram

Here in our Use Case diagram, we explained that camera will capture the real time video of the farm in regular time intervals. After that it extract the features and by using the algorithm dataset, it will identify the animal and its name. Then according to the name of the animal it will produce the sound according to the animal. Animal which entered the field, it receives the sound. In this way, we drive the animal away.

CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modelling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information. Class diagrams are the blue prints of your system or subsystem. Class diagrams are useful in many stages of system.

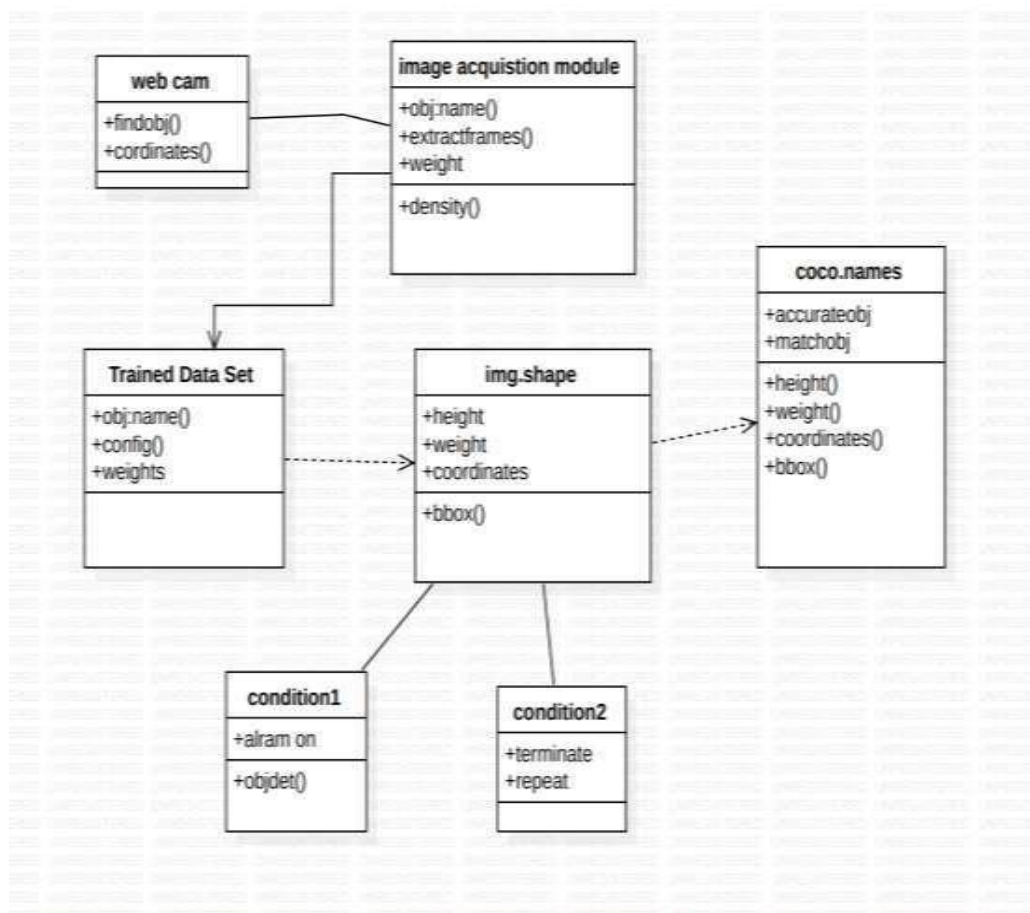


Fig 4.3 Class diagram

In this Class Diagram, we have various classes, methods and their attributes. In web cam, we have `findobj()` method which is used to find the object and coordinates. It is linked to the image module in which weights are calculated and from the trained dataset and **coco.names**, identifying animal name can be done and Sound is produced.

ACTIVITY DIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control. It is also termed as an object-oriented flowchart. It encompasses activities composed of a set of actions or operations that are applied to model the behavioral diagram.



Fig 4.4 Activity diagram

Here in our Activity Diagram, we first take the input data and analysis it. We compare it with the training dataset and test the input data. After the testing, we will get the result according to our input data that is sound is produced according to the animal that is detected

SEQUENCE DIAGRAM

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. The sequence diagram represents the flow of messages in the system and is also termed as an event diagram. It helps in envisioning several dynamic scenarios. It portrays the communication between any two lifelines as a time-ordered sequence of events, such that these lifelines took part at the run time. In UML, the lifeline is represented by a vertical bar, whereas the message flow is represented by a vertical dotted line that extends across the bottom of the page. It incorporates the iterations as well as branching. It is a construct of a Message Sequence Chart.

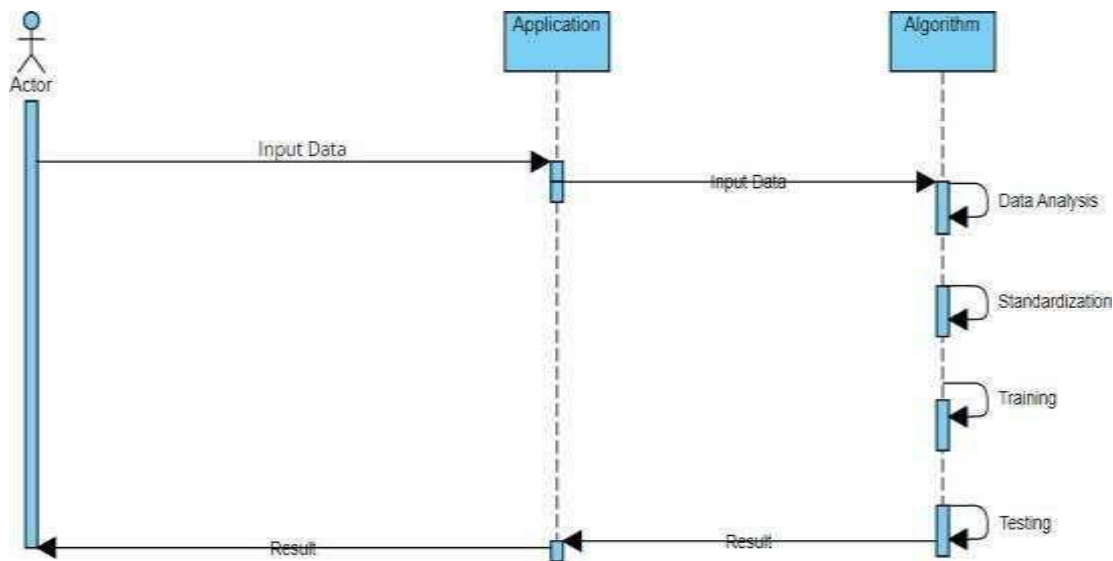


Fig 4.5 Sequence diagram

Here in our Sequence diagram, input data is passed to the application and then to the algorithm. In algorithm, data analysis is done and comparison of the input data to the training data will be done. After the testing, the appropriate results will be taken as the output. Here, in dataset, animal names are stored. After capturing the image which is taken as input, according to animal's name, sound will be produced to drive the animal away. This is shown in the sequence diagram that shown above.

5. IMPLEMENTATION

Coco.names Dataset:

A COCO dataset consists of five sections of information that provide information for the entire dataset. The format for a COCO object detection dataset is documented at COCO Data Format info – general information about the dataset. licenses – license information for the images in the dataset.

COCO dataset, meaning “Common Objects In Context”, is a set of challenging, high quality datasets for computer vision, mostly state-of-the-art neural networks. This name is also used to name a format used by those datasets.

YOLOv3 Algorithm:

YOLOv3 (You Only Look Once, Version 3) is a real-time object detection algorithm that identifies specific objects in videos, live feeds, or images. YOLO uses features learned by a Deep Convolutional Neural Network to detect an object. Versions 1-3 of YOLO were created by Joseph Redmon and Ali Farhadi.

The first version of YOLO was created in 2016, and version 3, which is discussed extensively in this article, was made two years later in 2018. YOLOv3 is an improved version of YOLO and YOLOv2. YOLO is implemented using the Keras or OpenCV deep learning libraries.

Open CV

OpenCV is a huge open-source library for computer vision, machine learning, and image processing. OpenCV supports a wide variety of programming languages like Python, C++, Java, etc. OpenCV will help you know about the Image-processing from Basics to Advance, like operations on Images, Videos using a huge set of OpenCV- programs and projects.

The CV component contains mainly the basic image processing and higher-level computer vision algorithms; MLL the machine learning library includes many statistical classifiers as well as clustering tools.

Computer vision is an approach to understanding how photos and movies are stored, as well as manipulating and extracting information from them. Artificial Intelligence depends on or is mostly based on computer vision. Self-driving cars, robotics, and picture editing apps all rely heavily on computer vision.

Human vision has a resemblance to that of computer vision. Human vision learns from the various life experiences and deploys them to distinguish objects and interpret the distance between various objects and estimate the relative position.

The main features to choose OpenCV

Specific OpenCV was designed for image processing. Every function and data structure has been designed with an Image Processing application in mind. Meanwhile, Matlab, is quite generic. You can get almost everything in the world by means of toolboxes. It may be financial toolboxes or specialized DNA toolboxes.

Speedy After doing some real time image processing with both Matlab and OpenCV, we usually got very low speeds, a maximum of about 4-5 frames being processed per second with Matlab. With OpenCV however, we get actual real time processing at around 30 frames being processed per second.

Efficient With OpenCV, we can get away with as little as 10mb RAM for a real-time application. Although with today's computers, the RAM factor isn't a big thing to be worried about. However, our drowsiness detection system is to be used inside a car in a way that is non-intrusive and small; so a low processing requirement is vital.

Deep Learning Approach

A deep approach to learning concentrates on the meaning of what is learned. That concentration may involve testing the material against general knowledge, everyday experience, and knowledge from other fields or courses. A student taking a deep approach seeks principles to organize information. In contrast a student using a surface approach tries to capture material in total, rather than understand it. An example is the student who busily copies down a diagram without listening to the explanation of it. As the above problem stated is still prevailing despite of all the methods taken, we approached the problem using deep learning to drive away the animals automatically. In our project, we used packages like Kera's and Play sound to do the pre-processing steps involved and to create. Here, the input is received from the CCTV (Closed Circuit Television), the code does the processing and prediction of the frames received from the camera and appropriate repellent sound is played to drive away the detected animal.

Object Detection

Object Detection is a computer technology related to computer vision and image processing that deals with detecting instances of semantic objects of a certain class (such as humans, buildings, or cars) in digital images and videos.[1] Well-researched domains of object detection include face detection and pedestrian detection. Object detection has applications in many areas of computer vision, including image retrieval and video surveillance.

When we look at images or videos, we can easily locate and identify the objects of our interest within moments. Passing on of this intelligence to computers is nothing but object detection- locating the object and identifying it. Object Detection has found its application in a wide variety of domains such as video surveillance, image retrieval systems, autonomous driving vehicles and many more. Various algorithms can be used for object detection but we will be focusing on YoloV3 algorithm.

Flow Chart

A flowchart is a type of diagram that represents a workflow or process. A flowchart can also be defined as a diagrammatic representation of an algorithm, a step-by-step approach to solving a task. First the image is taken as input for the YOLOv3 Algorithm and then it is configured with its configurations and weight. Then it is mapped with coco.names data set and the name of the object is obtained. If it is an animal extermination sound is produced otherwise next frame is fed as input.

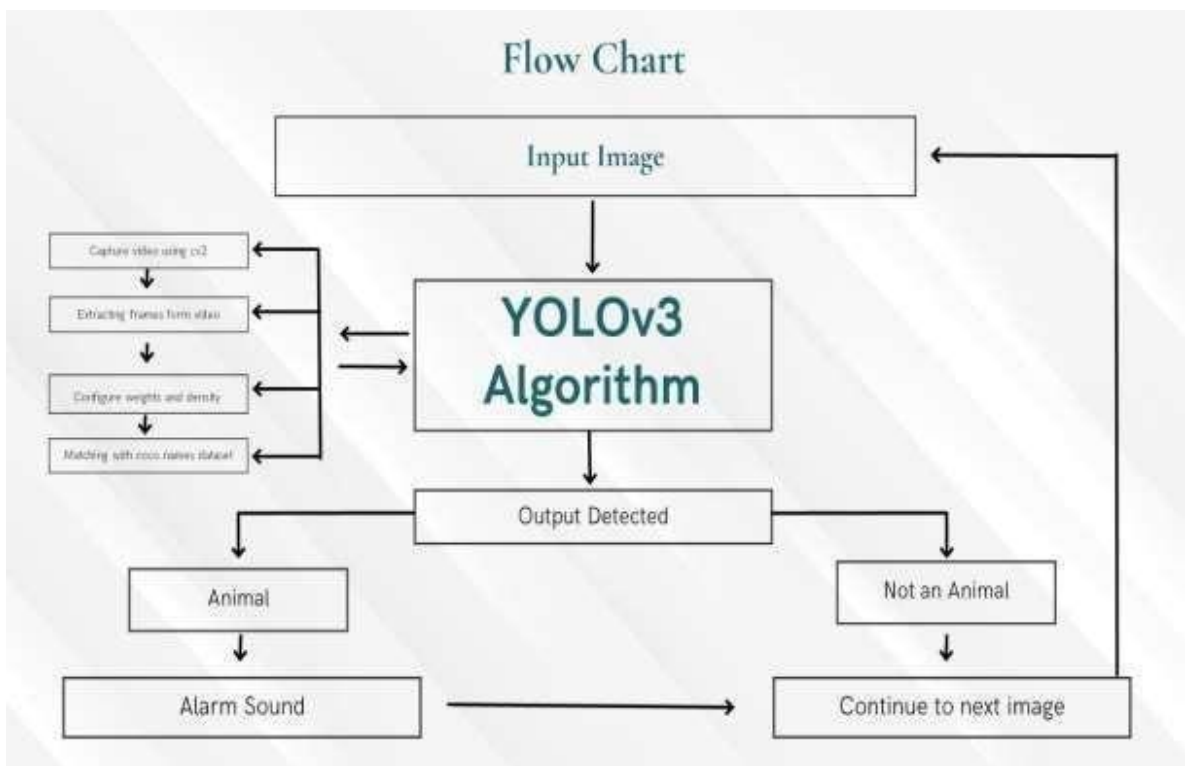


Fig 5.1. Flow chart of animal detection

5.2. Sample code

App.ipynb

```
import cv2

import numpy as np

import os

from twilio.rest import Client

# Twilio setup

account_sid = 'AC5a94e9f075402cb182a60ede72ec4422'

auth_token = '48a13bf1b3c7640287bc9080084bdf2e'

twilio_client = Client(account_sid, auth_token)

twilio_phone_number = '+19378073392'

alert_phone_number = '+919347243604'

# Load YOLOv3 model

modelConfig = 'yolov3.cfg'

modelWeights = 'yolov3.weights'

net = cv2.dnn.readNetFromDarknet(modelConfig, modelWeights)

net.setPreferableBackend(cv2.dnn.DNN_BACKEND_OPENCV)

net.setPreferableTarget(cv2.dnn.DNN_TARGET_CPU)

# Load class names

classesFile = 'coco.names'

classNames = []

with open(classesFile, 'rt') as f:

    classNames = f.read().rstrip('\n').split('\n')
```

```

# Initialize video capture (use webcam, change to video file path if needed)
cap = cv2.VideoCapture(0)

def send_sms_alert(animal):
    message = f'Alert: {animal} detected!'
    twilio_client.messages.create(
        body=message,
        from_=twilio_phone_number,
        to=alert_phone_number
    )

def find_objects(outputs, img):
    ht, wt, ct = img.shape
    bbox = []
    classIds = []
    confs = []
    for output in outputs:
        for det in output:
            scores = det[5:]
            classId = np.argmax(scores)
            confidence = scores[classId]
            if confidence > confThreshold:
                w, h = int(det[2] * wt), int(det[3] * ht)
                x, y = int((det[0] * wt) - w / 2), int((det[1] * ht) - h / 2)
                bbox.append([x, y, w, h])
                classIds.append(classId)

```

```

        confs.append(float(confidence))

indices = cv2.dnn.NMSBoxes(bbox, confs, confThreshold, nmsThreshold)

if len(indices) > 0:
    for i in indices.flatten():
        box = bbox[i]

        x, y, w, h = box[0], box[1], box[2], box[3]

        cv2.rectangle(img, (x, y), (x + w, y + h), (0, 0, 0), 2)

        cv2.putText(img, f'{classNames[classIds[i]].upper()} {int(confs[i] * 100)}%',
                    (x, y - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.9, (255, 255, 255), 3)

        detected_animal = classNames[classIds[i]]

        if detected_animal in ["horse", "elephant", "zebra", "giraffe"]:
            os.system("start C://Users//user//Desktop//mini_project//crackersounds.mp3")

            send_sms_alert(detected_animal)

        elif detected_animal in ["cat", "dog", "sheep"]:
            os.system("start C://Users//user//Desktop//mini_project//lionroarsound.mp3")

            send_sms_alert(detected_animal)

        elif detected_animal in ["cow", "bird", "mouse", "monkey"]:
            os.system("start C://Users//user//Desktop//mini_project//21-gun-salute-25444.mp3")

            send_sms_alert(detected_animal)

confThreshold = 0.5

nmsThreshold = 0.3

wht = 320 # Set the desired width and height for blob processing

```

```

while True:
    success, img = cap.read()

    if not success:
        break

    blob = cv2.dnn.blobFromImage(img, 1/255, (wht, wht), [0, 0, 0], 1, crop=False)
    net.setInput(blob)

    layer_names = net.getLayerNames()
    outputnames = [layer_names[i - 1] for i in net.getUnconnectedOutLayers()]
    outputs = net.forward(outputnames)

    find_objects(outputs, img)

    cv2.imshow('Webcam Feed', img) # Display the processed image in a window

    if cv2.waitKey(1) & 0xFF == ord('q'): # Break the loop if 'q' is pressed
        break

cap.release() # Release the webcam

cv2.destroyAllWindows() # Close all OpenCV windows

```

6. TESTING

6.1. Testing

Introduction

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

Developing Methodologies

The test process is initiated by developing a comprehensive plan to test the general functionality and special features on a variety of platform combinations. Strict quality control procedures are used. The process verifies that the application meets the requirements specified in the system requirements document and is bug free. The following are the considerations used to develop the framework from developing the testing methodologies.

Build the Test Plan

Any project can be divided into units that can be further performed for detailed processing. Then a testing strategy for each of this unit is carried out. Unit testing helps to identify the possible bugs in the individual component, so the component that has bugs can be identified and can be rectified from errors.

TESTING STRATEGY

Types of Testing

1. Integration Testing
2. Unit Testing
3. Functional Testing
4. System Testing
5. White Box Testing
6. Black Box Testing
7. Acceptance Testing

INTEGRATION TESTING

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

FUNCTIONAL TESTING

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manual.

Functional testing is centered on the following items:

Valid Input identified classes of valid input must be accepted. **Invalid Input:** identified classes of invalid input must be rejected. **Functions:** identified functions must be exercised.

Output identified classes of application outputs must be exercised. **Systems/Procedures:** interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

SYSTEM TESTING

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

WHITE BOX TESTING

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

BLACK BOX TESTING

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works. Unit Testing Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

ACCEPTANCE TESTING

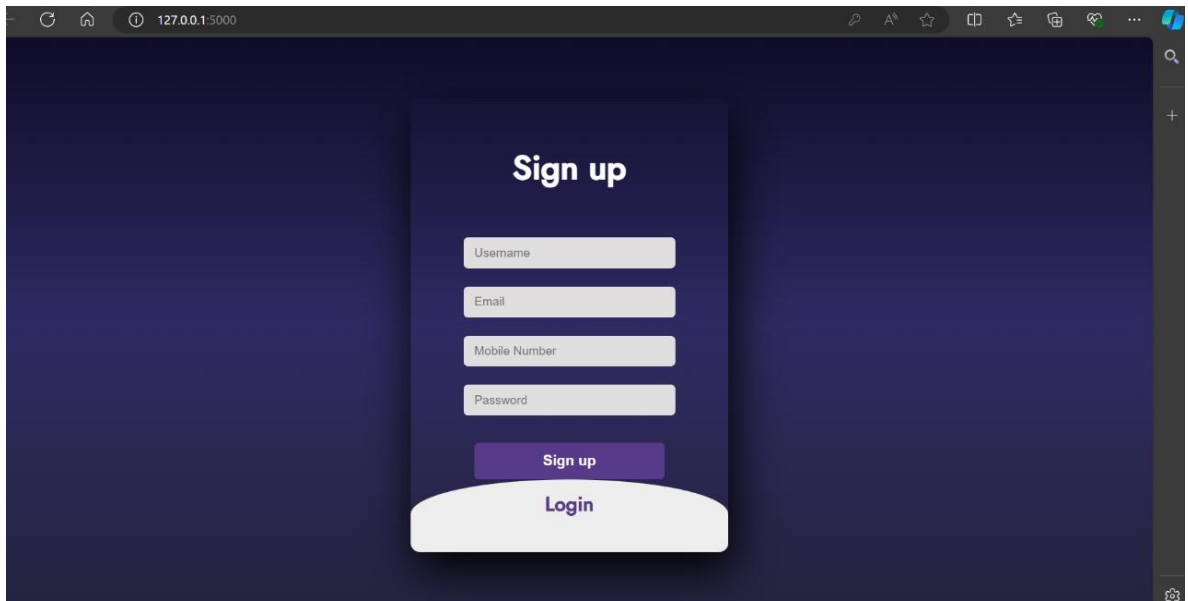
User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

6.2. Test cases

S.No	Test Case Name	Action	Expected	Results
1.	Camera Monitoring (successful)	Captured Image	Config, Weights	success
2.	Camera Monitoring (unsuccessful)	Not Captured Image	No Config, Weights	success
3.	Alarm (successful)	Validated Config, Weights	Sound Produced	success
4.	Alarm (unsuccessful)	Not Validated Config, Weights	No Sound Produced	success

Table 6.1 Test Cases

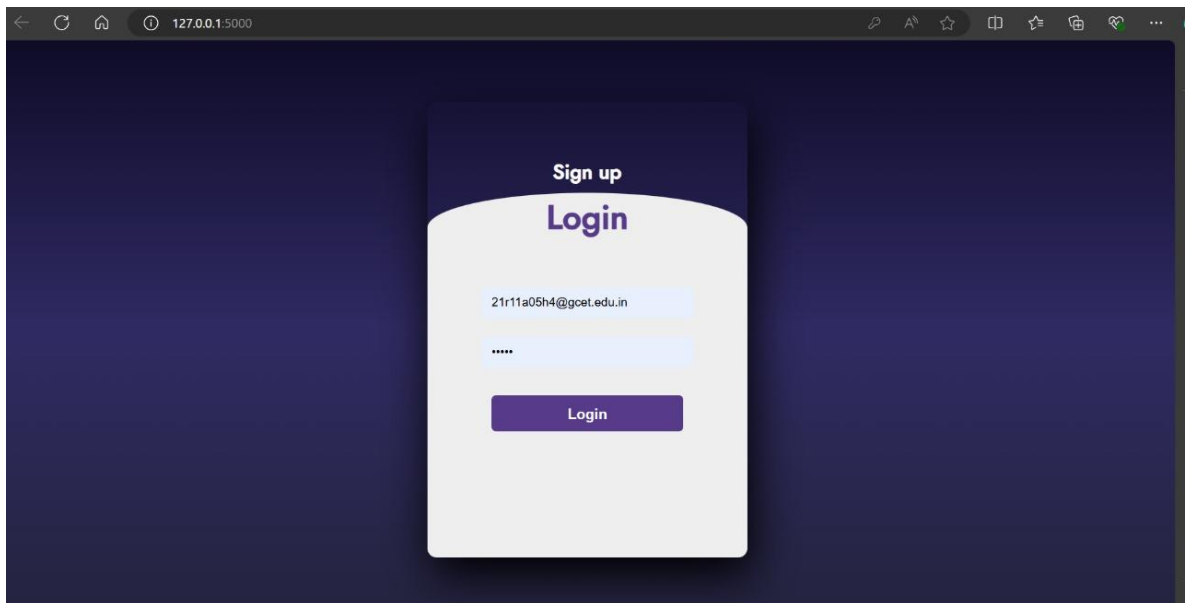
7. OUTPUT SCREENS



A screenshot of a web browser displaying a 'Sign up' form. The form is centered on a dark blue background. It features four input fields: 'Username', 'Email', 'Mobile Number', and 'Password'. Below these fields is a purple 'Sign up' button and a white 'Login' button. The browser's address bar shows '127.0.0.1:5000'.

Fig.7.1. Sign Up

This screenshot shows a registration form for the application that includes fields for a username, password, mobile number, and email address.



A screenshot of a web browser displaying a 'Login' form. The form is centered on a dark blue background. It features two input fields: a text field containing '21r11a05h4@gcet.edu.in' and a password field with masked characters '....'. Below these fields is a purple 'Login' button. The 'Sign up' button is visible above the 'Login' button. The browser's address bar shows '127.0.0.1:5000'.

Fig.7.2. Login

A login page is a user interface that allows registered users to access a application by entering their credentials, typically a username ora password. The main purpose of a login page is to authenticate users and ensure that only authorized individuals can access certain features or information within the platform.

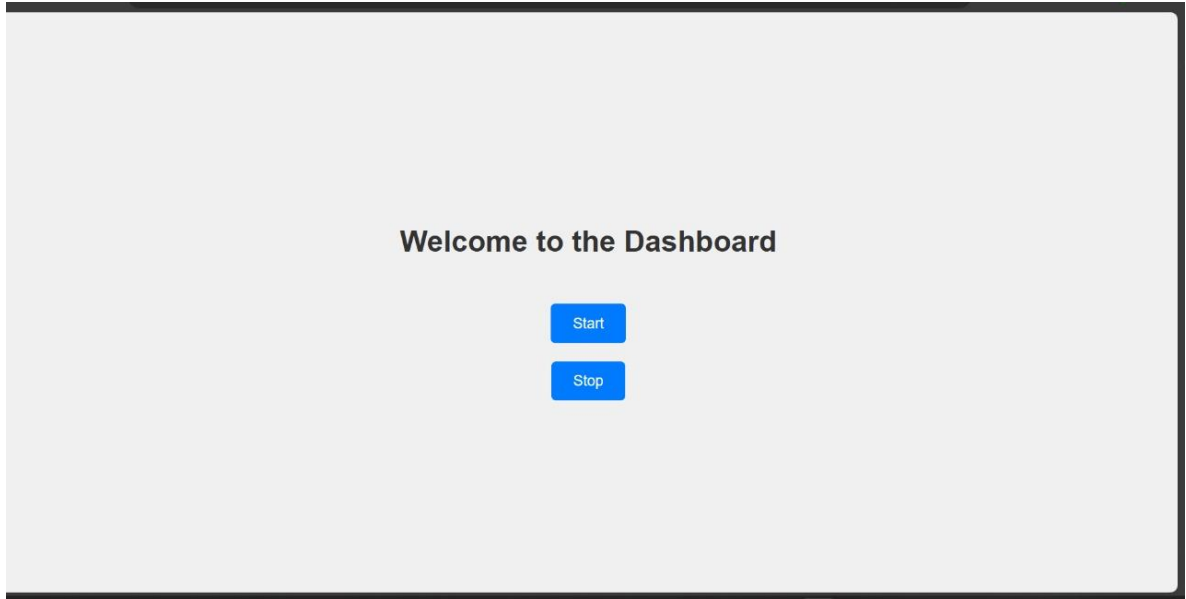


Fig.7.3. Dashboard

A dashboard designed for launching a webcam to view animals would serve as a centralized control panel, offering users a straightforward and intuitive interface to manage and monitor the webcam feed. This type of dashboard could be particularly useful for farmers, researchers to observe animals.

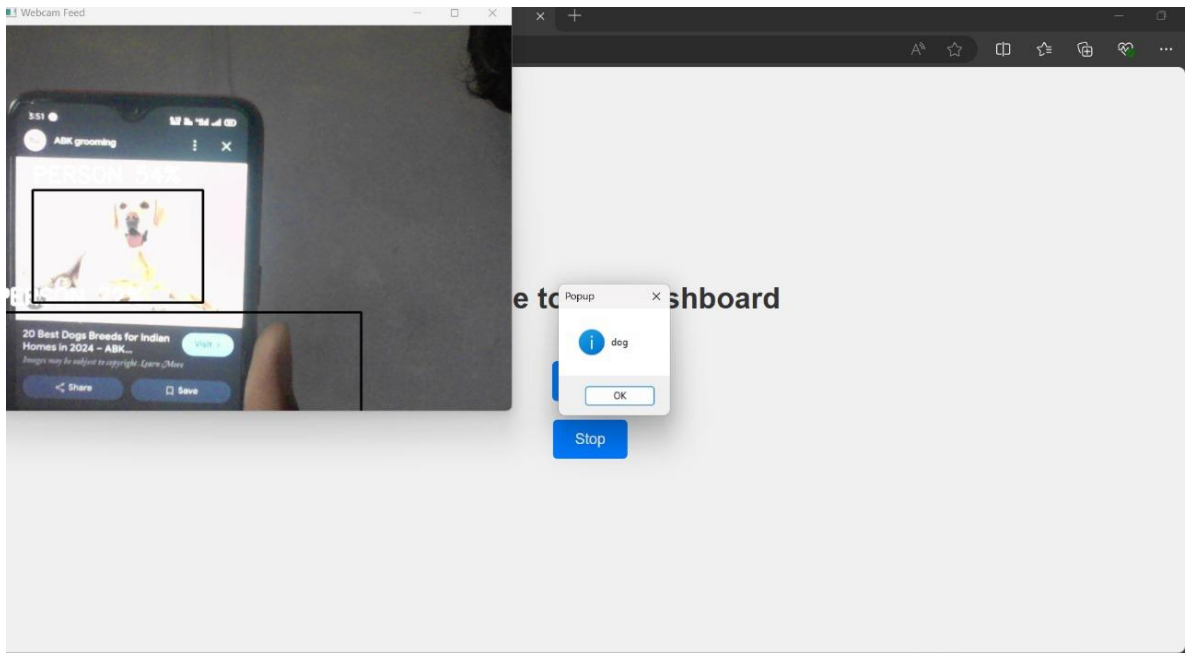


Fig.7.4. Monitor page

A monitoring page where a webcam displays live views of animals, automatically detects any animals present, and plays a sound to repel them. When an animal is detected, the system pops up a notification displaying the animal's name and sends an alert to the farmer.

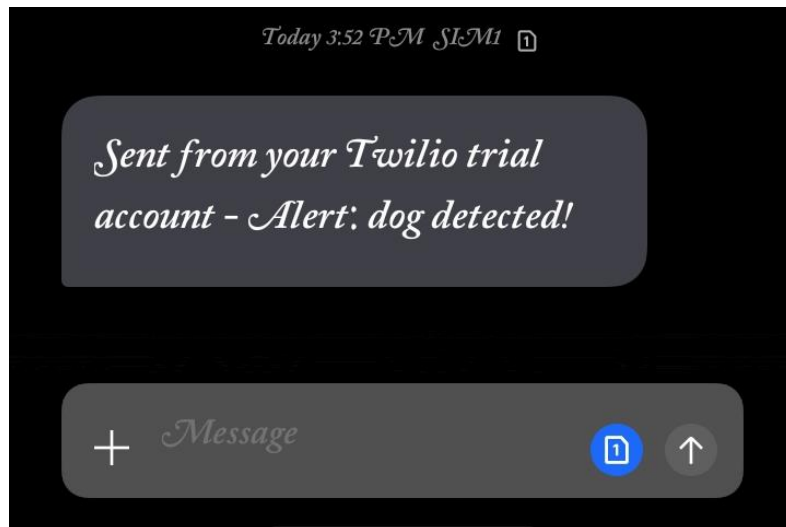


Fig.7.5. Alert in Farmer Phone

Alert message on farmer's phone from the application .

8. CONCLUSION

8.1. Conclusion

AI Based Scarecrow is a system that repels the wild animals that are trying to enter the field and exterminates them by playing the sound that they fear off. So, it can be concluded that we can recognize and reverse the animals before they enter the field by playing various repelling sounds and sending an alert message to farmer. The problem of crop vandalization by wild animals has become a major social problem in current time.

In other words, while utilizing his/her crop production, every farmer should be aware and take into consideration the fact that animals are living beings and need to be protected from any potential suffering. It requires urgent attention and an effective solution. By doing so, we reduce the crop loss and man power. This project is very useful and affordable to the farmer. The module will not be dangerous to animal and human being, and it protects farm. Thus, this project carries a great social relevance as it will help farmers in protecting their fields and save them from significant financial losses and will save them from the unproductive efforts that they endure for the protection of their fields. This ensures complete safety of crops from animals causing damage to it.

8.2. Further Enhancements

We are using an integrative approach in the field of Artificial Intelligence to overcome this. The goal of this work is to provide a repelling and monitoring system for crop protection against animal attacks. In our future work, we will extend the current functionalities of a model like increasing the dataset so as to achieve high accuracy and investigate the chance of incorporating the future of the model to other sectors.

It can be made into a robot and do the needful actions like moving hands and make it scare animals. It can be made to detect Human intrusion in order to stop robbery of crops.

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10. APPENDICES

Technical Specifications

This appendix provides detailed technical specifications for the hardware and software components used in the AI-based scarecrow project. It includes information on the cameras, edge computing devices such as Raspberry Pi and NVIDIA Jetson Nano, and ultrasound emitters. Additionally, it covers the software requirements, including the YOLOv3 algorithm and the playsound module, detailing the necessary versions and configurations.

System Architecture

In this section, the system architecture of the AI-based scarecrow is illustrated with diagrams depicting the data flow and interaction between components. The appendix includes detailed integration steps for combining camera feeds, YOLOv3 processing, and sound emission mechanisms.

YOLOv3 Configuration

This appendix details the YOLOv3 configuration files and parameters used for animal detection. It also includes information on the training data, preprocessing steps, and any adjustments made to optimize the model for the specific use case.

Sound Emission Protocol

Here, we provide a comprehensive list of sound files used for repelling different animal species and describe how the playsound module is configured for playback. This section also outlines the protocol for generating and triggering these sounds based on animal detection.

Experimental Results

This appendix includes tables and charts presenting the experimental results of the animal detection system. It covers detection accuracy metrics such as precision and recall, as well as data on temperature and power consumption of the edge computing devices during testing.

Case Studies and Field Testing

This section summarizes field tests conducted to evaluate the system's effectiveness in real-world conditions. It includes case studies highlighting different scenarios and the observed impact on crop protection.

Additional Resources

This appendix provides references to additional research papers and articles relevant to the technology used in the project. It also includes links to online tools and resources for further exploration.

11. PLAGIARISM REPORT

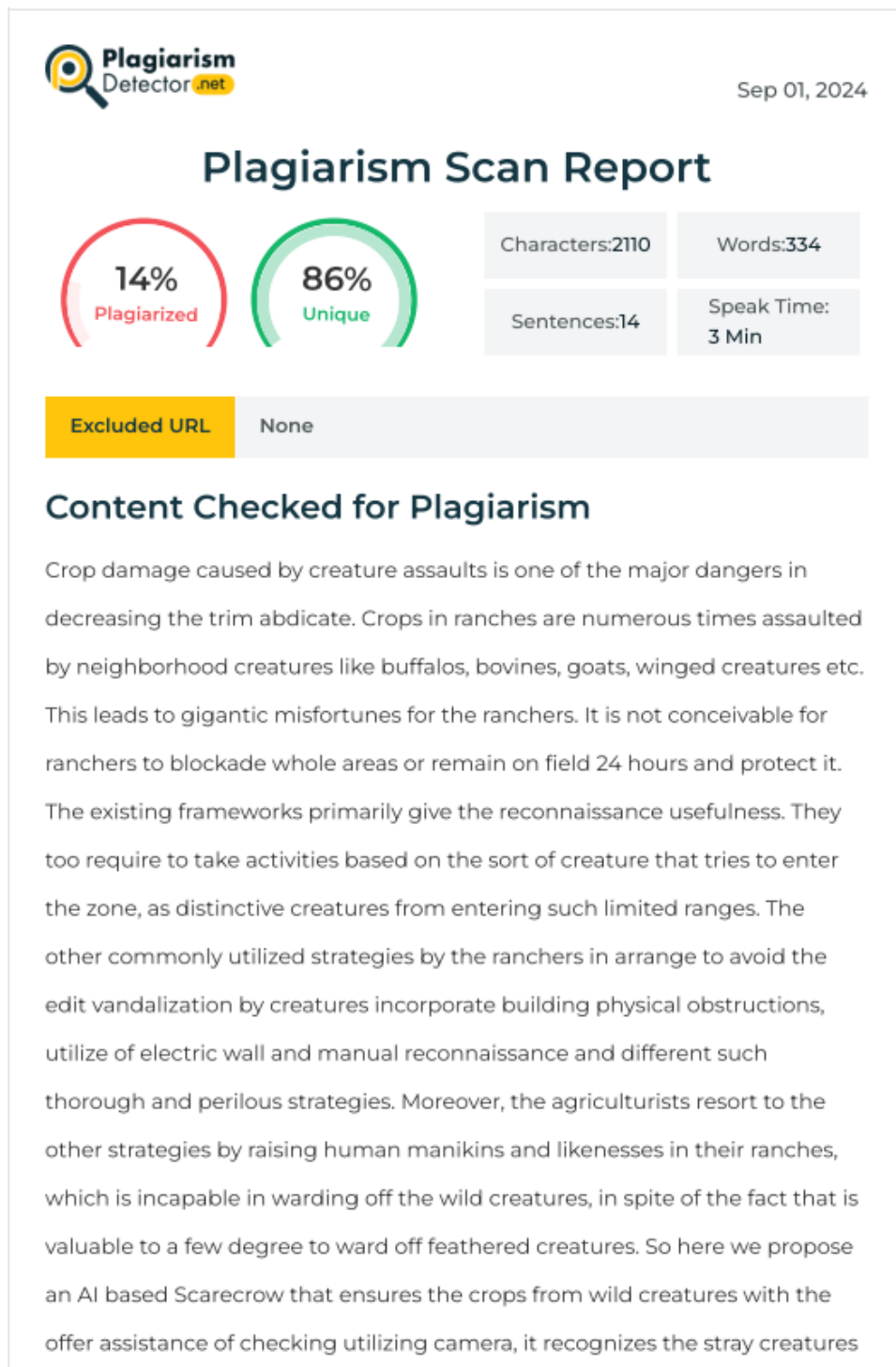


Fig.11.1. Plagiarism Report